

Preliminary Exploration of Sonification for Spatial Augmentation of Ball-Approach Dynamics and Timing in Tennis

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ABSTRACT

Open-skill sports require training methods that address perceptual-cognitive functions that influence action timing, in addition to movement technique. Sonification can support action timing by mapping continuous task dynamics to sound without visual overlays. We present a preliminary design exploration of sonification for real-time ball-position feedback in tennis. We implement an end-to-end system that estimates ball position using event-based vision and renders distance dynamics as continuous audio. We examine a 2x2 design space that crosses representation style (explicit tone vs. implicit wind) with frequency range (low: one octave vs. high: two octaves), reflecting a trade-off between change legibility and attentional demand. Pilot observations suggest that highly interpretable sonification can facilitate users' motor timing perception, whereas implicit designs may support attentional stability.

Index Terms: Spatiotemporal augmentation, Sonification, Tennis

1 INTRODUCTION

Open-skill sports such as tennis demand rapid adaptation in dynamic and uncertain environments [15]. In these settings, performance relies on perception-action coupling: the dynamic interplay between perception and action that supports real-time spatiotemporal anticipation, decision-making, and movement execution. In tennis, players must anticipate and initiate movement while tracking a fast-moving ball; effective play depends not only on movement technique but also on detecting changes in ball motion and translating them into spatiotemporally coordinated actions.

However, most training systems focus on form analysis and motor execution. In sports science and human-computer interaction, feedback systems quantify kinematics and support technique correction using computer vision, wearables, and virtual reality (VR) [4, 9, 13]. Some VR-based approaches also target perception [16], for example by visualizing ball spin in table tennis [17] or providing sonification for tennis returns that reflects timing demands [6]. However, perception in virtual environments can differ from real settings, which may limit transfer to on-field performance. Training support should therefore also target anticipation and decision-making and their coupling to action during live play, where movements must be executed under temporal constraints.

This gap motivates training approaches that support perception-action coupling during live play, yet visual feedback can interfere with performance. In open-skill sports, athletes must extract critical visual information (e.g., ball flight and opponent motion) while coordinating movement. However, vision is ill-suited to conveying fine temporal structure, and visual overlays can increase cognitive load by competing for limited visual attention [14]. These contexts therefore benefit from feedback that conveys continuous spatiotemporal dynamics without consuming visual resources.

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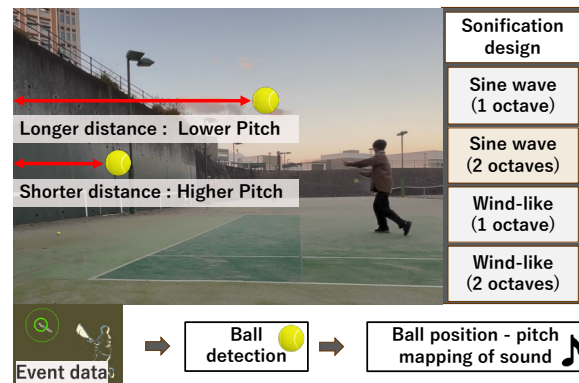


Figure 1: Augmented ball-position perception through real-time sonification. Our system tracks ball position using event-based vision and renders the ball's lateral position in the camera view as sound. Specifically, horizontal image position (distance from the left edge) is mapped to pitch, with larger distances producing lower pitch.

Sonification, widely used in motor learning [12], is a promising modality because it externalizes time-varying environmental information as sound, enabling access to fine-grained temporal changes through an auditory channel that can be integrated with visual processing and motor coordination [10].

In this work, we investigate sonification as an augmented-perception approach for stroke timing support during play [7]. Leveraging a low-latency event-based tracking system [8], we explore an auditory design space for ball-position feedback in tennis (Fig. 1). By mapping ball-distance dynamics to auditory pitch, the system externalizes spatial information into an auxiliary perceptual channel, supporting spatiotemporal perception without overloading visual attention. We define a 2x2 space based on representation style (explicit tones vs. implicit wind-like sound) and frequency range (one octave vs. two octaves). Pilot observations from a tennis wall-hitting task provide insights into the system's potential and inform our design direction.

2 RELATED WORK

Auditory cues contribute to timing judgments in sports, and audio-visual integration supports adding sound when visual attention is occupied. Prior work in ball sports indicates that impact- and motion-related sounds influence prediction and response, suggesting that auditory information supports anticipatory judgment [1]. Beyond discrete impact cues, work in music listening suggests that temporally unfolding auditory patterns can elicit prediction-based responses before an event [11]. In elite badminton, reaction times to sport-specific audio-visual stimuli were faster than to auditory-only or visual-only stimuli, indicating a benefit of multisensory integration in time-critical responses [5]. Together, these findings suggest that well-designed auditory patterns can facilitate perceptual anticipation and response in time-critical tasks; therefore, predictive sounds can support users' anticipation and action timing.

Sonification has been used to support motor learning and coordination. Prior research in sports and movement training has often mapped self-motion signals (e.g., rhythm and kinematics) to sound to support learning and proprioceptive awareness [2, 12]. In contrast, fewer studies have examined continuous sonification of externally driven dynamics during natural play, where latency, robustness, and attentional compatibility are critical constraints. Immersive-environment studies have used spatialized sound to encode direction and distance for guidance [3]. In ball sports, however, auditory augmentation should also convey temporal structure and remain compatible with perception–action coupling during live play, making real-world evaluation essential. Building on spatial guidance work, we explore sonification design choices for ball-position information to support spatiotemporal coordination in tennis. To this end, we combine event-based ball tracking with a 2×2 factorial manipulation of sonification style and pitch range, and we collect qualitative feedback to characterize trade-offs between interpretability and interference.

3 SONIFICATION DESIGN AND IMPLEMENTATION

Our objective is to help players at an intermediate skill level or below extract timing information from ball motion and develop a stable execution rhythm through perceptible approach cues. To this end, we implemented an end-to-end pipeline that integrates event-based ball tracking with real-time sonification of ball position, using an event camera. Fig. 2 shows the experimental setup and the camera view. In our wall-hitting setup, the event camera was positioned court-side; in the camera view used for this experiment, the wall appears on the left and the player appears on the right. By externalizing horizontal ball-position dynamics as sound, the system supports motor timing during execution. The auditory feedback complements vision, enabling coordinated use of auditory and visual cues without visual overlays. Leveraging the high temporal precision of auditory perception, we use sonification to represent the temporal structure of ball approach.

In the detection phase, we used an event camera positioned court-side to estimate ball position with low latency by exploiting luminance events; its asynchronous intensity-change reporting enables robust sensing of fast-moving objects without motion blur. We tracked ball position using a lightweight, model-based approach that treats the ball as an isolated high-speed object over short temporal intervals, enabling real-time operation without learning-based models [8]. The resulting position stream was passed to the sonification module, where distance changes were rendered as a continuous temporal cue during action execution.

In the sonification phase, all conditions share the same mapping rule. We mapped the ball’s horizontal position (left-to-right in the camera view, with the wall on the left and the player on the right) to pitch within a predefined frequency range, such that rightward motion (i.e., the return trajectory from the wall toward the player, referred to as the “approach” phase in our wall-hitting setup) produced a lower pitch. Low-span range designs restrict this range to one octave, whereas high-span range designs extend it to two octaves. Sound levels are adjusted so that loudness remains approximately constant across pitch.

We compared four sonification designs defined by two factors: representation style and frequency range. Representation style contrasts an explicit, readily interpretable mapping with an implicit mapping intended as a background cue, while frequency range controls the magnitude of spectral change for a given ball displacement. For the implicit condition, we used a wind-like sound, implemented by shifting the peak frequency of a noise spectrum to convey position. We chose this wind-like design because it is closer to everyday environmental sounds and may therefore be more naturally accepted as an implicit background cue. Crossing these factors yields a 2×2 design space (explicit tone vs. implicit wind; low one-octave

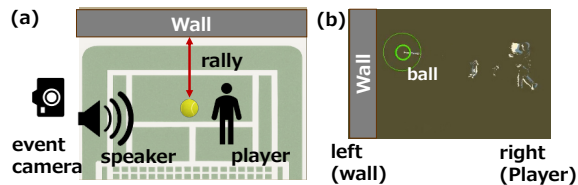


Figure 2: Experimental condition: (a) system setup; (b) camera view

span: 200–400 Hz vs. high two-octave span: 200–800 Hz). This design probes a practical trade-off: larger spectral modulation can enhance the perceptibility of fine-grained motion information, but increased saliency may raise attentional demands during visually guided action. We adopted a linear mapping to maintain a uniform relationship between distance and spectral change. This isolates the effects of representation style and modulation range.

The system was run on an ASUS ROG Strix G16 (2025) G614 equipped with an NVIDIA GeForce RTX 5070 Laptop GPU, and audio was delivered through a Soundcore Motion X600 portable speaker, which was placed court-side near the player. In our preliminary study, we processed the event stream in frames at 70 fps.

4 PRELIMINARY STUDY AND QUALITATIVE RESULTS

We conducted a preliminary on-court study to examine how sonification design choices affect perceived timing support during hitting. We collected qualitative feedback on how players interpreted sound changes during ball approach and how the sonification influenced attention during stroke execution. The study involved three participants, two male and one female, each with more than three years of tennis experience. Two participants were intermediate and one was intermediate to advanced, and one participant was also an author.

Participants provided informed consent prior to participation. Participants performed standard tennis wall-hitting while the system tracked the ball and generated real-time auditory feedback from the detected position stream. Each session began with wall-hitting without auditory feedback as a warm-up, followed by sequential exposure to each sonification condition for approximately one to two minutes, after which qualitative feedback was collected through brief semi-structured interviews.

Result Qualitative feedback revealed differences across players. Intermediate participants (P1, P2) preferred explicit tonal feedback with larger pitch ranges, noting that salient spectral changes made ball distance clear and provided rhythmic cues aligned with stroke execution. Smaller-range tonal conditions were described as ambiguous and harder to interpret. P2 noted that wind-like noise evoked the impression of poor weather (e.g., strong wind), making it uncomfortable and harder to integrate for timing. An intermediate-to-advanced participant (also an author) preferred smaller-range wind-like sound, stating that it synchronized with body movement and supported sustained concentration without drawing attention. This preference may reflect the author’s prior familiarity with the wind-like mapping, which reduced the effort required to interpret the feedback. In contrast, larger spectral changes were described as distracting. They added that the feedback supported a stable hitting pace and rhythm rather than explicit prediction of ball timing. Overall, these reports suggest that sonification can support hitting performance but involves trade-offs among spatiotemporal interpretability, rhythmic support, and attentional compatibility.

5 CONCLUSION AND FUTURE WORK

This work presented a preliminary study of real-time ball-position sonification for timing support in tennis wall-hitting. We examined

whether sonification can augment stroke execution by rendering ball-approach information through sound while players maintain visual tracking and movement coordination. Although the sample size was small and the findings are based on subjective reports, the results suggest that auditory augmentation can support spatiotemporal coordination during repeated strokes.

This study has several limitations. The small sample size ($N = 3$), including one participant who was also an author, prevents statistical generalization and may introduce bias. In addition, presenting conditions in a fixed sequence may have produced order or carryover effects. Finally, this pilot relied on short-term subjective reports rather than objective measures of timing or performance. Although participants did not report problematic latency, we did not quantify end-to-end system latency or temporal jitter, for example audio buffering and device-related delays. Moreover, tracking was not always stable, occasionally leading to brief dropouts in the sonification when the ball was not reliably tracked. Future studies should report these metrics and examine how latency, variability, and tracking robustness affect perceived timing support and performance. To validate these observations and assess potential training effects, we will conduct larger controlled experiments linking quantitative outcomes such as swing-initiation timing relative to ball approach, impact-timing consistency, and movement stability to sonification parameters, including information content and saliency. Cognitive workload should be assessed using standard instruments such as NASA-TLX, and coach feedback may help refine sonification design for training and skill development. We also plan to test the approach across other ball sports with different tempo and flight dynamics (e.g., badminton or table tennis) to clarify its generality and sport-specific design requirements.

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